CANONICAL

CIO guide to multi-cloud operations

How to choose a cost effective cloud architecture

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Introduction

The question of whether cloud computing is an advantageous way for businesses to optimise costs and become more agile is no longer up for debate.

Yet leveraging the cloud to the greatest effect, and maximising return on investment, is more complicated than merely making the decision to opt into cloud computing. The cloud can take many forms – from public and private to hybrid and multi-cloud. Some clouds are delivered as managed services, while others offer infrastructure, often combined with services on top. Clouds can be constructed using a diverse range of technologies, including bare metal servers, virtual machines, application containers, system containers and more, whilst many organisations use multiple types of clouds and cloud services at the same time - known as a multi-cloud strategy.

Due to the wide range of adoption tactics, determining exactly which workloads to migrate to the cloud, which type or types of cloud architecture to adopt and how to facilitate the cloud journey is a complex process.

It is also an extremely important process. Choosing the right type of cloud environment, and the right technologies for constructing it, is essential both for operating a business at maximum efficiency and keeping your organisation's tools, processes and personnel as agile as possible. No one type of cloud architecture of infrastructure is ideal for everyone. Developing the right cloud strategy requires evaluating your business needs and aligning them with the different types of cloud solutions available.

In the following pages, you'll find objective discussion of the various approaches to cloud computing that are available today, an explanation of how the cloud intersects with technologies like containers and serverless computing, and the factors that organisations should weigh when deciding which route to the cloud is the best match for their needs.

Choosing which cloud architecture (or combination of architectures) to adopt, and which infrastructure technologies to embrace, are not the only important decisions that need to be made when planning for a modern cloud strategy. Businesses must also gain familiarity with the different types of tools available for helping to build and manage a cloud.

Defining cloud architectures and infrastructure technologies

Let's start by defining exactly what we mean when we discuss different types of cloud architectures and services. While some of the material in this section may seem quite basic to readers who are already deeply experienced with cloud technology, it is important to clarify this information because some of these topics are not always discussed in a consistent way. The following definitions of different types of cloud architectures establish a baseline for identifying and evaluating various cloud strategies.

If you are not yet using the cloud in any way, your infrastructure runs on-premises in a traditional manner, with little or no use of virtualisation or distributed environments. Conventional on-premises computing is the paradigm that predominated before the widespread adoption of cloud computing starting about a decade ago.

In order to make the transition from traditional computing to the cloud – and to decide whether some of your workloads should remain on-premises – you first need to understand the main types of cloud architecture that are available today. They include the following:

- **Public cloud.** In a public cloud environment, compute, storage and other infrastructure resources are provided as a service by an external provider. Importantly, however, management and support for the software that runs using those resources is not typically part of public cloud offerings. Major public cloud providers include Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform, although a list comprising of all the companies that offer a public cloud would be much longer. While the most common use case for the public cloud involves running virtual servers in a public cloud environment, a variety of other public cloud services are available, from private bare-metal servers, to hosted Docker containers, to serverless computing like Lambda Functions.

- **Private cloud.** A private cloud provides organisations with compute, storage and other resources that can be accessed over the network and that are reserved exclusively for them. The main difference between private and public cloud architectures is that in the private cloud model, organisations do not share cloud infrastructure with other organisations. Private cloud also provides the advantage of being able to determine exactly how the cloud environment is configured, and which technologies are used to build it. Private clouds can be built using platforms like OpenStack, an open source solution for combining clusters of infrastructure to form a cloud. They can be built using on-premises infrastructure or in private hosted environments such as rented data centre space through an Equinix or Digital Realty. Nowadays, internal IT functions are also able to offer a private cloud with SLAs that closely aligns with the needs of the business. The SLAs can relate to various factors such as performance, or adherence to specific data compliance regulations.

Managed cloud. While public and private cloud platforms do not generally include support services for setting up or maintaining servers and applications that run in the cloud, managed cloud offerings offer this across private, public and hybrid cloud infrastructures. Managed cloud can be the easiest on-ramp to implementing a cloud, due to it removing the challenges posed by using internal resources. However, it typically comes with higher direct costs in the form of service fees. That said, the total cost of ownership of a managed cloud may be lower because it eliminates the need to employ a team of sysadmins.

Hybrid cloud. A hybrid cloud architecture is one that combines the usage of a number of public and private cloud services with orchestration between the two platforms. Hybrid cloud can also be managed, with, typically, an external company managing IT operations across the cloud environments. For example, an organisation with a hybrid cloud architecture may use a public cloud service to host frontend applications, while relying on a private cloud for storing sensitive data that cannot be moved to public cloud storage. In practice, medium and large organisations with multiple cloud resources often operate under a hybrid cloud model.

Multi-cloud. A multi-cloud architecture is the form of hybrid cloud towards which most of the industry is converging, with 79% of businesses already using more than one provider, according to a survey by 451 Research, commissioned by Microsoft. It is typically made of one private cloud (self-operated or managed) and two public clouds.

In addition to these different approaches to cloud architecture, a diverse set of infrastructure building-blocks are available for building modern clouds. They include:

- **Virtual machines.** As noted above, the classic scenario for running a workload in the cloud centers on using traditional virtualisation software (such as a KVM or VMware hypervisor) to create a virtual server (VM) that runs in the cloud. Dozens of virtual servers typically run on the same bare-metal host server, whose resources are shared by users of the virtual servers.

- **Bare-metal servers.** Some clouds also include bare-metal servers. Bare-metal servers can provide better performance for certain types of applications, but they generally cost more and may not be as scalable as other cloud-based resources.

- **Process containers.** Process containers provide isolated environments that host individual processes or services. They have smaller footprints than virtual machines and are therefore more efficient and scalable. Docker, which debuted in 2013, is the most popular process container platform today, although older process container frameworks exist. While process containers gained popularity through consumption in the public cloud, they're increasingly being run on premise for cost and privacy reasons.

- **System containers.** Container technology can also be used to host entire applications and not just individual processes. To the application it creates an environment that looks and behaves like a typical operating system/VM, but are in reality portable environments much lighter than a traditional VM. These types of containers, which are available from platforms such as LXD, are called system containers. Because system containers require much fewer resources than do traditional hypervisors, system containers are ideal for building "dense" cloud environments where many virtual operating systems run on a single bare-metal host server. LXD system containers can support environment densities 14.5 times greater than those of traditional virtual

machines, while also starting and responding much faster. LXD is a "really fast hypervisor" that lets you operate at container speeds. LXD takes the speed and latency of containers and brings them to the hypervisor world. A LXD container is designed to give you full 'machine' system functionality, not just a single process. You can run Docker and RunC inside LXD to mix container types.

- **Serverless computing.** Serverless computing (sometimes also called serverless functions) provides an execution environment where small pieces of code can be run on demand. Because cloud-based serverless computing services require users to pay only when they are actively using the service, rather than having to maintain a virtual server on an ongoing basis, they are ideal for completing tasks that require high processing power for only a short time, such as image resizing. Strategic use of serverless computing can save tens of thousands of dollars per month in infrastructure costs, as this story from Postlight illustrates.

These various infrastructure components are not either/or propositions Mostmodern clouds are built using a combination of these technologies and various cloud services. Organisations should take advantage of the options available for hosting different types of workloads. For example, it may make sense to run some workloads in virtual servers, some in containers and some using serverless functions, depending on the performance, scalability and cost requirements associated with each type of workload.

The infrastructure technologies you choose may also be determined in part by the type of cloud architecture you embrace. For instance, if you use a public cloud and wish to avoid the "noisy neighbor" problem that can result from sharing hosting resources with other virtual machines running on the same physical server, you may wish to run workloads on a private cloud-based server instead.

Choosing the right cloud architecture and infrastructure

Planning the right cloud strategy for your business requires evaluating the various options available and aligning them with your needs. Below are the major advantages and disadvantages associated with each type of cloud architecture, as well as insights about the infrastructure considerations associated with each one.

What is public cloud?

The public cloud tends to be the first option to which organisations turn when planning to migrate workloads from an on-premise to a cloud-based environment. In many cases this makes good sense. Public cloud platforms provide a number of benefits, including:

- **Minimal setup time and maintenance costs.** In the public cloud, users do not have to purchase or set up the physical infrastructure that hosts their applications. (They do, however, have to provide their own support for the software that they install on their infrastructure; for more on this requirement and alternative options, see the discussion of managed cloud below.) The capital expenses associated with public cloud are therefore low, although ongoing operating expenses are generally not.

- **Launch applications globally within minutes.** Tapping into a global footprint of data centres means that businesses can reach new audiences around the world in minutes. A key benefit being lower latency, due to the application being hosted in a location much closer to the customer.

- **On-demand access to compute and storage resources.** In the public cloud (as in most other types of cloud configurations) compute, storage and other resources can be accessed whenever users need them, with minimal setup required. Public cloud providers also tend to roll out innovative new services at a rapid pace, offering users an ever-expanding set of options for hosting workloads and consuming public cloud resources.

- **The ability to scale resource consumption virtually without limit.** Public clouds provide virtually unlimited pools of resources. This allows organisations to scale quickly from hosting a handful of services in the cloud to running thousands. This type of instant scalability is not available in most on-premise infrastructures because expanding on-premise environments require the purchase and installation of new hardware, which is a time-consuming and costly process.

- **Choice of operating systems.** Public clouds provide the flexibility to choose from a range of different host operating systems -- including different types of Linux distributions, as well as multiple versions of Windows. For example, AWS supports more than 250 different operating system choices, with the most popular being Ubuntu. In addition, because most public cloud providers offer operating system images that are preconfigured to run on their infrastructure, users can spin these operating systems up quickly with minimal manual configuration.

- **Optimised software stack.** Public cloud vendors like AWS, Azure, or Google work with operating systems and hardware vendors to create optimised versions of the software stack for their hardware, providing smaller image sizes, improved performance and lower boot time. Canonical for example works with all the above players to create optimised versions of Ubuntu for their respective clouds.

- **Lower staffing and requirements.** Because the public cloud eliminates the need to buy, configure and maintain infrastructure, it also, by extension, lowers staffings costs, or allowing the existing IT team to focus on tasks that bring greater value to the business than 'keeping the lights on', or carrying out low value tasks. The public cloud allows organisations to access a very large infrastructure with a small IT support team.

The major public clouds also offer a range of different options for hosting workloads. From serverless functions to containers to virtual servers, users can access almost any type of infrastructure solution on the public cloud without having to install and maintain the service themselves.

Yet the public cloud is not a panacea. The public cloud is not the best fit for all types of workloads, whilst the requirements of some organisations may make the public cloud a less practical choice. The characteristics of these organisations are as follows:

- **Large infrastructure requirements.** Organisations with large or steady infrastructure needs can typically meet them in the most cost-efficient way by building their own on-premise or private cloud infrastructures and maintaining the large support teams required to manage them. They may still take advantage of public cloud services to serve purposes that their own infrastructures cannot support, such as peaks in demand.

- **Data privacy and sovereignty requirements.** Regulatory compliance and security concerns can complicate the migration of some types of workloads to the public cloud, as can data sovereignty requirements that restrict the movement of data between different political jurisdictions. In these cases, private cloud is often a better solution because it provides users more control over how and where data is stored.

- **Minimal technical expertise.** While the public cloud simplifies the setup and management of infrastructure in many ways, public cloud resources nonetheless require more than a negligible amount of expertise to use. Organisations with very little infrastructure and architecture technical expertise on staff may be better served by using a managed cloud service to maintain their public cloud environment.

The aforementioned reasons are often factors in the decision to choose private, hybrid, or multi-cloud architectures, rather than going all-in with public cloud. In summary, the public cloud is an excellent option for making workloads more scalable and agile, and it is often the first place that organisations look when planning to migrate to the cloud. However, the public cloud is not the right fit for all organisations and workloads.

What is private cloud?

In certain situations, the private cloud offers a better architecture. Private cloud is advantageous in the following scenarios:

- **Your organisation has applications that can't run in the public cloud.** Some types of applications are not compatible with the public cloud. For example, a data analytics application may not always work well in the public cloud because data offloading from on-premise data sources can take too long over the public Internet. In this case, a private cloud with private network connections will offer better performance.

- **Compliance and data privacy regulations prevent an application from running in the public cloud.** Compliance policies such as PCI DSS and HIPAA regulate the ways in which data can be stored and moved. While these policies do not prevent data from being stored in the public cloud under any circumstances, complying with the policies can be more challenging on public cloud platforms because users do not have full control over the way resources are managed. For example, PCI DSS imposes requirements on access control related to data stored in virtual servers, but public cloud users typically do not have the ability to control how the public cloud provider manages its employees' access to its users' data. In situations like these, private cloud can simplify the process of remaining compliant.

- **High performance is a priority.** As noted above, an organisation with a private cloud does not share its hosting resources with other organisations. While individual users within the organisation may still share resources, private clouds, in general, provide better performance because resources are not shared.

- **Data transfer costs are a concern.** In the public cloud, organisations typically pay not only for each node or virtual machine they run, but also for the data they transfer. Organisations that move large amounts of data may therefore be better served by a private cloud because in the private cloud model, they own the network and do not have to pay for each gigabyte of data that they transfer over the network.

- **You want flexibility in choosing infrastructure technologies.** Not all public clouds offer all types of infrastructure technologies. Some may only support certain types of virtual machine hypervisors, for example. Others only allow you to use application container with specific orchestration tools. This lack of choice constricts your ability to be agile. This is not the case in a private cloud. Platforms like OpenStack are capable of supporting almost any type of infrastructure, giving the user the freedom to pick and choose precisely which technologies are used to construct the cloud environment, and which services are available on it.

- **Organisations have a growing infrastructure.** For organisations whose infrastructure is expanding and is expected to grow to a large size, private cloud is typically more cost-effective than public cloud in the long run. However, whilst the last few years have seen enterprises consume more public cloud services, many have been hit by spiralling costs, thanks in part to evolving cloud requirements. In a recent report by 451 Research it was found that Canonical's managed private OpenStack offering, BootStack, delivers private cloud with a TCO that matches public clouds.

As noted above, private cloud is not without drawbacks. The total cost of ownership of a private cloud has in the past been deemed as being greater than that of an equivalent public cloud, when managed internally by an organisation's own staff and where the organisation's infrastructure needs are too small to make a private cloud cost-effective.However, when managed externally, the TCO of a private cloud can be equal to that of a public cloud deployment.

Private clouds do have the drawback of not always being as elastic as public clouds because expanding private clouds when their maximum capacity is reached requires acquiring and setting up new hardware, a process that demands time and capital.

Finally, private clouds require organisations, in most cases, to support their own infrastructure. The set-up and maintenance burden of private clouds can be higher than on public clouds because on unmanaged private clouds, organisations have to provide their own operations and support for both the cloud infrastructure and the applications and services that it hosts. However, a fully managed private cloud solutions such as Canonical's BootStack is capable of providing a production private OpenStack cloud in three weeks, see below.

Still, the private cloud option is often an ideal solution for achieving the agility and flexibility of the public cloud while solving the technical and compliance issues that organisations may face in the public cloud. Private clouds can also be a more cost-efficient option for organisations with large infrastructure or high data-transfer requirements.

What is managed cloud?

If the privacy, performance and cost-optimisation features of the private cloud are attractive, but you lack the resources to set up and manage your own private cloud, managed cloud may be the best fit. In addition to being easier set up, managed cloud offers several other key benefits:

- **Low "time-to-cloud."** Managed cloud offerings minimise the time needed to set up and start using a private cloud. A managed cloud offering like Foundation Cloud Build can be tailored to your organisation's needs and can be up and running in just days, along with a professional support team to manage it. In contrast, setting up a private cloud can take weeks or months, and acquiring the skilled admins required to manage it could take as many as six months.

- **No staffing costs.** Managed clouds do not require organisations to maintain a large in-house support staff. Management tasks are instead entirely outsourced. This saves not only staffing costs, but also the time and effort required to recruit and onboard new employees, as well as to retain skilled engineers. Just as importantly, it allows your existing IT engineers to focus on tasks that are at the core of your business -- like developing and maintaining business-critical applications -- rather than spending their time operating a cloud.

- **Easy access to expertise.** Taking advantage of newer infrastructure technologies like containers or serverless computing on a public or private cloud can be difficult because not all engineers are familiar with these technologies, and those on your staff may not be capable of supporting them. However, with a managed cloud plan, you gain much broader access to IT expertise. A managed cloud support staff can help you implement and maintain whichever type of infrastructure you choose.

- **Choice of managed services:** Canonical's managed offering, BootStack is a fully managed cloud service to help your organisation find the fastest path to a production private OpenStack cloud. With this offering, and Foundation Cloud Build - a fixed-price cloud with a proven reference architecture - it is possible to deploy an OpenStack private cloud in just two weeks, with two engineers, anywhere in the world. Once deployed, BootStack utilises high levels of automation to ensure TCO akin to the public cloud, with Canonical taking responsibility for the remote management and 24/7 availability of the OpenStack. An SLA covers uptime and performance, whilst hardware and software are both proactively monitored and scaled on demand with additional racks. Under this model the organisation can monitor every machine in the OpenStack cloud and be trained to take over control, with Canonical offering training and a handover process with on-site cloud reliability engineers to ensure a smooth handover.

What is hybrid cloud?

Organisations can deploy different workloads on different types of cloud infrastructure concurrently, creating a hybrid cloud architecture. Hybrid cloud architectures are often the best fit for organisations wishing to split the difference between public and private cloud offerings across a variety of technologies.

- **Choosing the type of hybrid cloud** that is right for your business, and exactly how to approach it, requires determining which types of workloads are best served by which cloud architectures, then arrange them accordingly. For instance, you may wish to build a small managed cloud for hosting specialised workloads that cannot be managed internally, while using public cloud services to host the rest of the applications.

Modern APIs make it possible to divide workloads across multiple types of infrastructure in quite granular ways. For example, the storage service for a web application might run in a private cloud (in order to meet data compliance requirements), while the public-facing web frontend is hosted in the public cloud (where the application has maximum scalability), with a network API connecting the two services.

Dividing workloads between different types of clouds according to a hybrid model helps not only to work around technical or regulatory challenges, but also to optimise costs. It allows organisations to run certain workloads in the type of cloud that is the least expensive for those particular workloads. For example, an organisation may use a private cloud for the majority of its infrastructure, but take advantage of serverless computing services available in the private cloud for running compute-intensive operations. This approach would allow the organisation to maintain a smaller private cloud, which costs less to acquire and manage, while still enjoying access to virtually limitless on-demand compute resources in the public cloud.

- **What is a hybrid cloud?:** It is also important to understand that a hybrid cloud architecture is not defined by the use of different types of infrastructure technologies at the same time. In other words, using virtual machines and Docker concurrently, for example, does not amount to a hybrid cloud. Each of these infrastructure technologies could run in a public cloud, a private cloud or a managed cloud at the same time. However, if one of them runs in one type of cloud and the other in another type, then they form a hybrid cloud.

- **When not to use a hybrid cloud:** Of course, hybrid cloud only remains cost-effective when each type of cloud environment that is used to build the hybrid cloud sees sufficient use. A hybrid cloud strategy fails if, for example, it leads to an organisation's private cloud resources being underutilised because too many workloads are moved to the public cloud. In a scenario like this, it would be better to focus on one type of cloud.

- **Cost-efficiency:** In cases where an organisation can truly benefit from deploying significant workloads to more than one type of cloud, a hybrid architecture makes most sense. A hybrid strategy provides maximum flexibility and cost-efficiency when executed in the right way, particularly for organisations whose workload requirements are diverse and can be best served by a combination of cloud architectures. To put some numbers on it, Cloud Technology Partners found average enterprise TCO savings for financial services of 42% from the adoption of hybrid cloud.

What is multi-cloud?

A multi-cloud architecture can be considered as a relative of the hybrid cloud, typically made up of one private cloud, operated either by the enterprise or managed by another party, and two public clouds. This architecture allows for greater flexibility than other models when it comes to optimising for cost and operating applications in their most optimal environment.

The multi-cloud architecture is quickly being seen as the de-facto cloud strategy for enterprises, with 79% of businesses working with more than one cloud provider, and 29% working with four or more cloud or hosting providers, and the adoption shift makes sense - as highlighted by the following benefits.

- **Vendor lock-in:** Originally considered one of the primary reasons for adopting a multi-cloud strategy, the desire to not be tied to one cloud vendor remains a driver for many organisations, although a wider range of factors can now be considered more significant influences. There remains value in not putting all your eggs in one basket, businesses remain fearful of having all their data in one provider's cloud infrastructure, in case a vendor pulls the plug or there are service issues.

- **Maintaining workload portability:** Although avoiding vendor lock-in is a key benefit of the multi-cloud model, making that a reality requires workload portability and avoiding proprietary APIs and applications. In essence, this means having two or more systems or components that are able to exchange information and to use it. This requires APIs that are able to talk to each other. The Cloud Standards Customer Council highlights in its report, 'Interoperability and Portability for Cloud Computing: A Guide Version 2.0' that portability can be broken down to the application and to the data. For the cloud, it is moving one cloud service to another cloud service, and on the data side, moving data among cloud and on-premises systems - typically using an API. Unfortunately, many APIs are proprietary and offer little to no standardisation with others, for example the AWS serverless compute service, Lambda, is not portable with Google Cloud Functions or Microsoft Azure Functions.They may have similarities, but they operate differently. Therefore, it is important to closely assess cloud services and APIs, ensuring that proprietary technology is not used without careful consideration when embarking on a multi-cloud journey.

- **Reliability:** An important factor for all organisations, given the necessity of an always-on infrastructure, always being up and running is the de-facto expectation. In a multi-cloud model it's possible for another cloud to take over should the primary cloud suffer any downtime, in essence serving as a failover solution.

- **Price-competitive:** The multi-cloud model hands the power back to the customer, removing the ability for the vendor to overcharge and underperform and handing enterprises a freedom of choice that allows them to maximise their ROI by choosing the most cost-efficient solution. Cost optimisation is one of the biggest drivers of multi-cloud adoption (according to a BMC survey).

- **Optimal application environment:** Embracing a mix of public and private platforms provides not only an ROI advantage over other models, it also promotes the hosting of the right applications in the right environments and the use of the best tool for the job, rather than sole reliance upon what a single vendor is able to offer. The reality is that some vendors have better tools for certain jobs than others, so the ability to pick and choose services from different vendors allows you to have the best available, not just what your provider currently offers. Choosing data centre regions or availability zones that are closest to the end user also helps to minimise latency, enabling applications to perform optimally for the end user.

- **Regulatory compliance:** Frequently changing regulatory demands requires an infrastructure architecture that is capable of meeting that change. Being tied to one vendor for either public or private cloud can result in unnecessary restrictions being placed on the organisation, potentially impacting how the organisation operates, regulations such as the German Bundesdatenschutzgesetz (BDSG), or Federal Data Protection Act. Other European nations also have extremely strict data protection laws, Russia for example requested that Twitter store data from Russian users on servers hosted in the country. Frequently these data protection and sovereignty laws require an enterprise to keep data in certain geographic locations for data sovereignty reasons. To ensure that this does not become a hindrance, organisations have sought to adopt a multi-cloud model to select services and providers from different data centre regions/ availability zones.

Canonical's cloud expertise

Canonical, whose software and service offerings cover virtually all cloud configurations and architectures, developed this whitepaper to help businesses determine which type of cloud architecture and which types of underlying infrastructure technologies are right for their organisation.

Canonical has not only witnessed, but also helped to drive, IT revolutions that have occurred over the past decade. From the explosion of Linux and open source into the commercial realm, to the rise of cloud computing, to the advent of microservices and containers over the past several years, Ubuntu and other Canonical platforms and services continue to play a key role in enabling technological change. From fuelling innovation by making open source software available to people everywhere.

To the advent of OpenStack as the de-facto choice for private cloud.

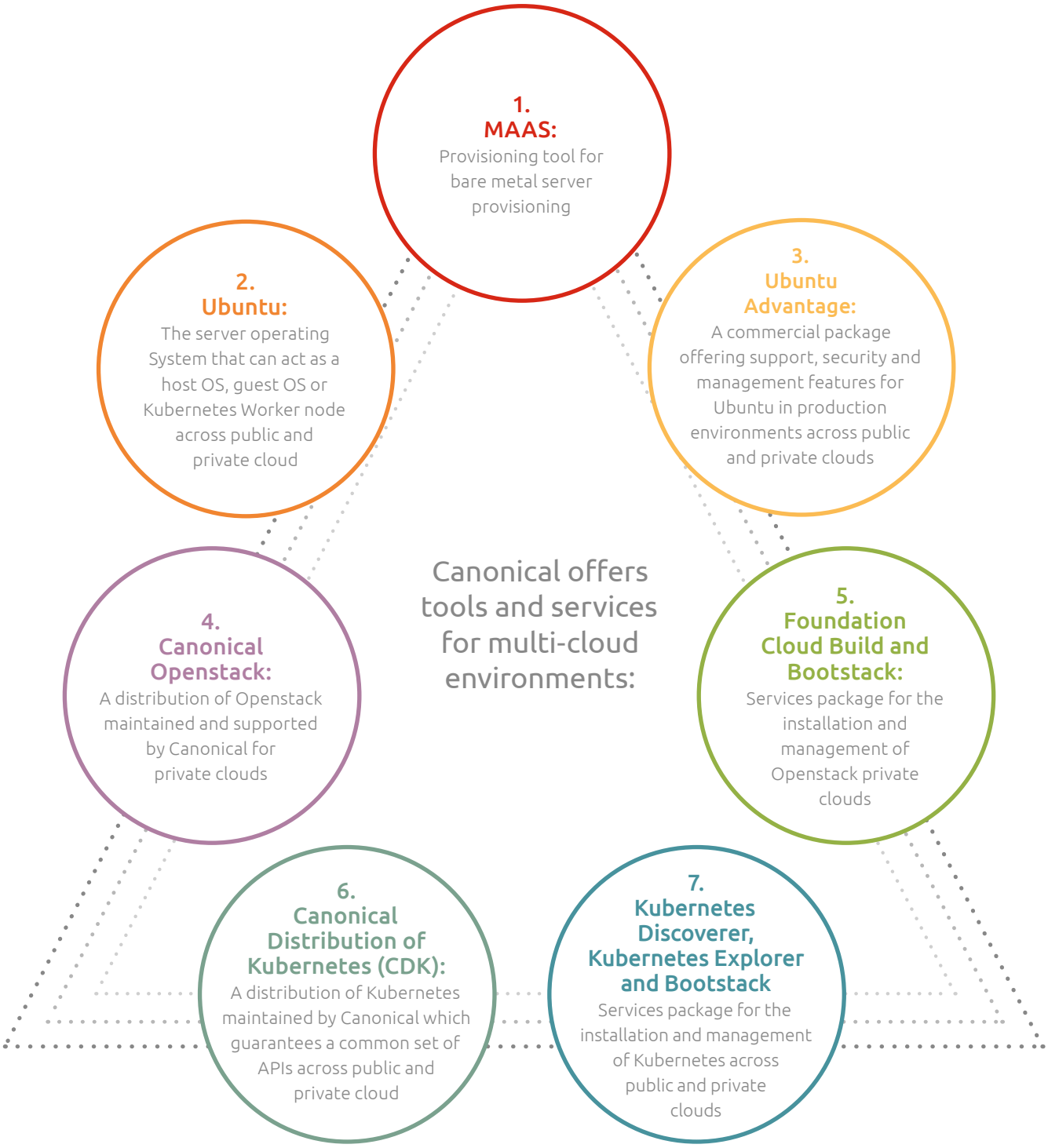
The evolution of containers and Kubernetes.

Canonical continues to push the industry for widespread change, to adopt the best methods and models for the best outcome, both technically and financially for the end user.

Canonical tools to help build the right cloud

This whitepaper has served to lead readers through the various cloud industry offerings, to empower them to be able to identify and pursue the cloud strategy that best fits their needs.

Now that the variety of cloud deployment architectures are clear, it's important to discover the kinds of tools that are available to help with the design, build, and management of clouds of varying types from Canonical.



Operating system software

Canonical's flagship product, Ubuntu, is an open source operating system that is widely used on all types of cloud architectures. It is popular both as a guest OS, whereby you run your application, and as a host OS, on which it is possible to put your VM, container, or where you can build OpenStack or Kubernetes infrastructure. Ubuntu is by far the most common operating system in the public cloud. Ubuntu can be combined with OpenStack to build private cloud environments of virtually unlimited scale. Ubuntu can be used to host virtual machines, containers, Kubernetes and all other major open source virtualisation and clustering platforms.

Cloud infrastructure

OpenStack, which traces its roots to a software project at NASA, is the preeminent open source platform for building cloud infrastructure today. OpenStack is an Infrastructure-as-a-Service (IaaS) platform that integrates a group of bare-metal servers (even if they are servers of varying types) into a cloud environment to make compute, storage, networking and other resources available on demand. Adoption rates of OpenStack deployments highlight that it is most commonly built as a private cloud, over a managed private or a public cloud, although these are also available and running in production use cases around the world.

OpenStack can support a variety of different service options -- from virtual servers to containers, and even bare-metal servers. While OpenStack requires Linux host servers, it can support guest virtual machines based on any Linux distribution, as well as modern versions of Windows.

Canonical supports OpenStack clouds that are built on Ubuntu, the world's most popular Linux-based open source operating system. In addition, Canonical offers a fully managed implementation of OpenStack as part of the BootStack managed cloud. Ubuntu powers 55% of OpenStack in production.

Cloud hypervisors and containers

Canonical both helps to develop and support two different types of open source hypervisor that can host virtual operating systems in a cloud environment. Simply put, a hypervisor is a process that separates a computer's OS and applications from the underlying hardware. A container, which everything at Google runs off, is a way of packaging application's code, configurations, and dependencies into building blocks for the purpose of delivering greater consistency, operational efficiency, version control at scale.

Canonical's products and support in these fields consist of:

Kernel-based virtual machine (KVM): KVM is a VM that uses the Linux kernel to emulate hardware and host virtual machines. KVM is supported on all modern versions of Ubuntu.

LXD: This is a system container. Rather than emulating hardware, LXD shares a host server's processes and file system with guest operating systems in order to create virtual environments that function like virtual machines but consume much fewer resources. LXD is ideal for helping to build clouds that require large numbers of virtual server-like environments to run on a single bare-metal host server.

Kubernetes / multi-cloud container orchestration

Containers are a technology that allows the user to divide up a machine so that it can run more than one application (in the case of process containers) or operating system instance (in the case of system containers) on the same kernel and hardware, and in so doing maintain isolation between these workloads.

Although similar to virtual machines in terms of the purpose, containers differ in several ways, primarily due to containers providing a way to virtualise an OS so that multiple workloads can run on a single OS instance. In VMs, it is the hardware that is being virtualized to run the multiple OS instances. This helps containers to be much lighter than VMs, megabytes rather than gigabytes, and much faster to start.

Kubernetes is an open-source software orchestration manager. Kubernetes is required for the management of containers, allowing the user to automate the deployment, scaling and operations of process containers across a set of machines, ensuring that the entire operation can be done without human/DevOps intervention. Kubernetes (from the Greek 'κυβερνήτης' meaning 'helmsman') was developed by Google, its design having been heavily influenced by Google's 'Borg' project – a similar system used by Google internally on which most of its infrastructure is run. Kubernetes has since been donated to the Cloud Native Computing Foundation, a collaborative project between the Linux Foundation and Google, Cisco, IBM, Docker, Microsoft, AWS and VMware.

Also known as K8s, the technology has emerged as a leading choice for organisations looking to build their multi-cloud environments, thanks to the widespread adoption of its APIs. AWS Elastic Container Service for Kubernetes, Google Kubernetes Engine and Azure Container Services have all adopted the technology.

In a multi-cloud world, Kubernetes can act as a way for organisations to create a common environment on which to run their containerised workloads.

Canonical's position with this technology is unique in that Ubuntu powers the majority of Kubernetes in public clouds. Canonical also offers a clean upstream distribution of the Kubernetes which can be deployed across public and private cloud.

Although an important element of the move to multi-cloud architectures, Kubernetes isn't the only tool required, serving more as one piece of the puzzle and the tools below helping to fill in the gaps.

Cloud orchestration and management

For provisioning and managing cloud environments, Canonical offers several tools:

Juju, an orchestration tool that automates the process of deploying and managing infrastructure and applications across a large number of cloud servers.

Metal-as-a-Service, or **MAAS**, an end-to-end cloud automation solution that transforms bare-metal servers into cloud-based virtual machines in order to construct a private cloud.

Landscape, a management tool for administering and monitoring multiple Ubuntu-based servers.

Ubuntu Advantage (UA) is the professional support package from the experts at Canonical, helping organisations around the world to manage their Ubuntu estates.

BootStack is a fully managed cloud service from Canonical and the fastest path to a production private OpenStack cloud. The service is aimed at allowing your business to focus on the business, whilst Canonical takes care of building and running your OpenStack cloud.

Foundation Cloud Build is a consulting offering through which Canonical designs and deploys your OpenStack on your premises

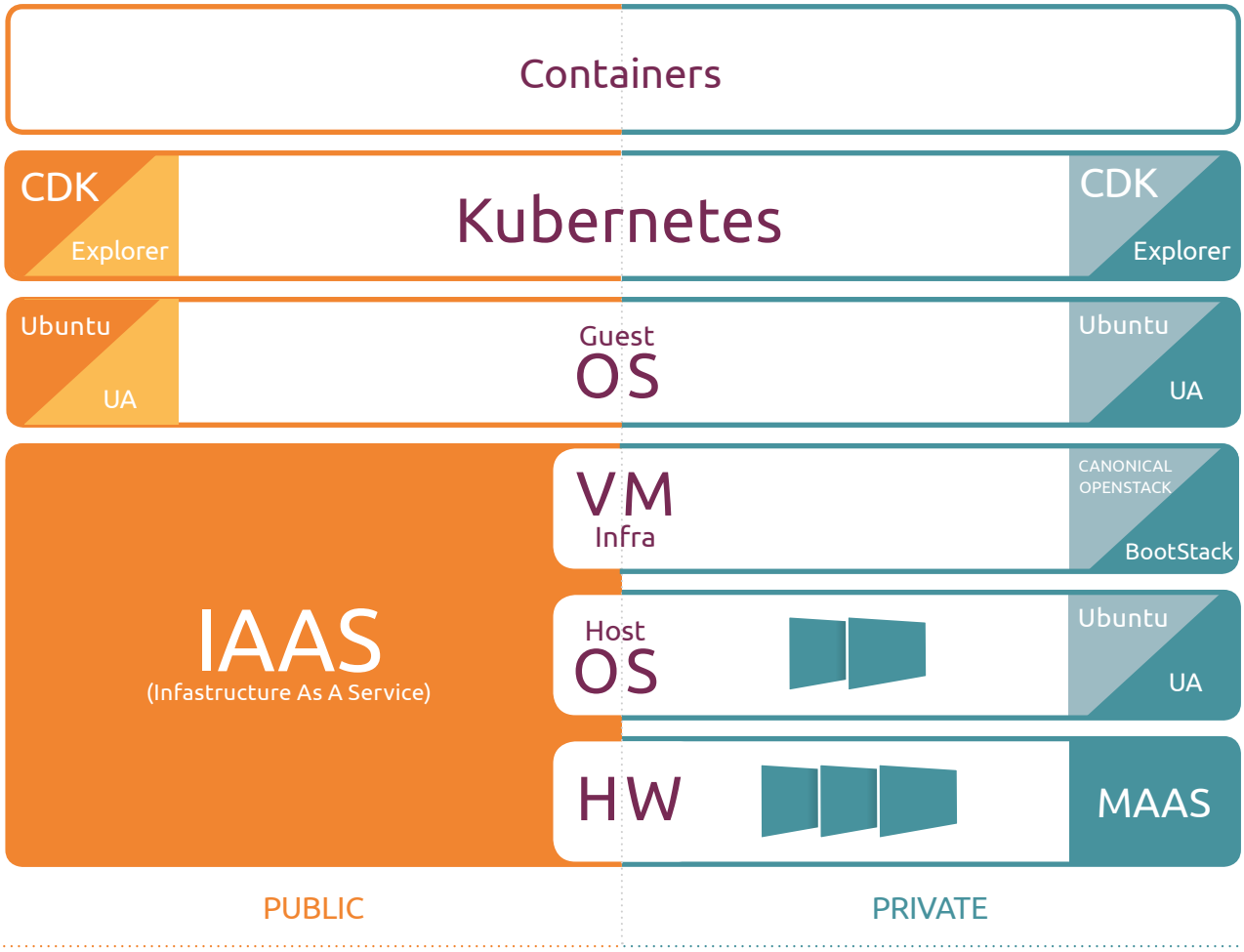
Kubernetes Explorer is a service package offered by Canonical that is designed to help your business launch its Kubernetes strategy. This package offers three-day training on Canonical Kubernetes and tooling, helping you ramp up your Kubernetes skills and get you ready to deploy in your own environment.

The value of these technologies is not limited strictly to the cloud. These tools can also help to set up and manage on-premise servers and, in some cases, even workstations.

For organisations seeking to make the most of the cloud, tools like these are essential. By automating tedious tasks and maximising visibility, these tools help to operate clouds at peak efficiency, no matter what type of cloud architecture chosen. And because they all have open source roots, they maximise your organisation's agility by allowing you to avoid vendor lock-in and other constraints of proprietary software.

The Canonical Stack

In summary, Canonical offers a wide portfolio of products to build a multi-cloud, illustrated below.



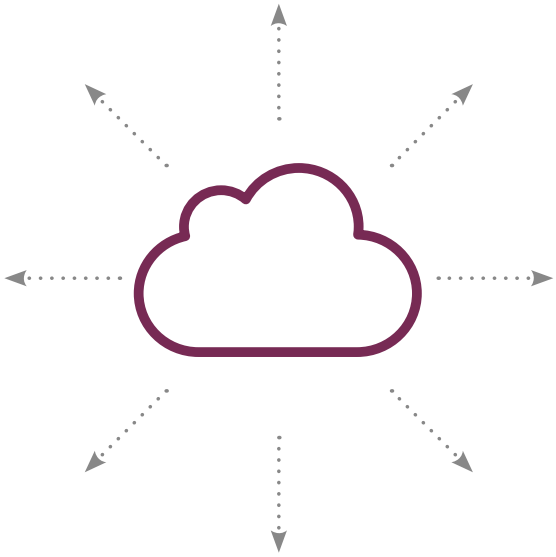
Conclusion

There is no one-size-fits-all type of cloud. Nor is the cloud necessarily the best option for hosting workloads in all situations.

Instead, choosing whether to use the cloud, and developing the right approach for architecting your cloud, requires identifying your business and technology needs and aligning them with the options available.

The push to begin organisational transformations may begin with competitive pressure, changing economics and the need for cutting costs, or a need to enable innovation through the modernisation of an IT estate. Whatever the driving force, there are a multitude of flavours of cloud to choose from.

Adoption of cloud technologies can and should be flexible to fit the desires of the customer, whether that is at a small or large scale, with one vendor or many.



As an open source company that develops an array of technologies for building infrastructures of all types, Canonical can help you to plan and execute the ideal cloud strategy. Whether the business need is for a professionally supported operating system like Ubuntu to power virtual servers in the public cloud, a production-ready OpenStack distribution, a hypervisor such as LXD for creating virtual environments or a fully managed cloud service via BootStack, Canonical delivers the software and services you require.

If you want to learn more and to discover how Canonical can help your business with its cloud journey, please click here

Further reading:

eBook: How to escape StuckStack and profit from your OpenStack investment

Report: Insights into multi-cloud economics

Datasheet: Foundation Cloud Build

Overview: The number one OpenStack

Datasheet: Managed Cloud from Canonical

Overview: Ubuntu in the Public Cloud

White paper: Why Ubuntu for Containers

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